Representation theory

Definition: A representation $G \longrightarrow GL(V)$ is irreducible (simple) if V has no proper non-trivial G-invariant subspaces. It is decomposable if it is equivalent to $\varrho_1 \oplus \varrho_2$, where ϱ_1, ϱ_2 are non-trivial representations.
Note: Decomposable $\Longleftrightarrow$ Complementary G-invariant subspaces of dimensions >0.

Lemma (Schur): Let A be a $\mathbb{C}$-algebra. Then
1. S, T simple A-modules, $f: S \longrightarrow T$ a module homomorphism $\Rightarrow f \equiv 0$ or iso.;
2. S simple, $\dim S < \infty$, $f: S \longrightarrow S \Rightarrow f = \lambda \mathrm{id}_S$.

Proof:
1. $f \not\equiv 0 \Rightarrow \operatorname{Ker} f \lneq S \Rightarrow \operatorname{Ker} f = 0$; $0 \neq \operatorname{Im} f \leq T \Rightarrow \operatorname{Im} f = T$.
2. f has an evector s with $f(s) = \lambda s \Rightarrow \operatorname{Ker}(f - \lambda) \neq 0 \Rightarrow f - \lambda \equiv 0$. $\square$

Corollary: S simple, $\dim S < \infty \Rightarrow \operatorname{Hom}_A(S,S) = \mathbb{C}$.

Prop: A $\mathbb{C}$-algebra, $z \in Z(A)$, S simple A-module $\Rightarrow z \cdot s = \lambda_z s \ \forall s \in S$
Proof: The map $s \longmapsto z \cdot s$ is a module homomorphism. $\square$

Corollary: If A is commutative then its simple modules are 1-dimensional.
But there might be simple ∞-dim modules, eg the regular $\mathbb{C}(t)$-module

Definition: For $x \in G$, let $x^G = \{gxg^{-1} : g \in G\}$.
Note: $Z(\mathbb{C}G) = \{z \in \mathbb{C}G : gzg^{-1} = z \ \forall g \in G\}$.
Definition: The conjugacy class sums are $\sum_{c \in x^G} c \in Z(\mathbb{C}G)$.
Note: The conjugacy class sums form a basis for $Z(\mathbb{C}G)$. In particular,
$$\dim Z(\mathbb{C}G) = \#\text{ conj. classes in } G.$$

Definition: Let M, N be $\mathbb{C}G$-modules. Then $\operatorname{Hom}_{\mathbb{C}}(M,N)$ is a $\mathbb{C}G$-module via
$$g \cdot \alpha : m \longmapsto g\alpha(g^{-1}m)$$
Definition: If X is a $\mathbb{C}G$-module then
$$X^G = \{x \in X : gx = x \ \forall g \in G\}$$
Note:
$$\operatorname{Hom}_{\mathbb{C}}(M,N)^G = \operatorname{Hom}_{\mathbb{C}G}(M,N).$$

Let
$$e_1 = \frac{1}{|G|} \sum_{g \in G} g \in Z(\mathbb{C}G).$$
Note that
$$M^G = e_1 M.$$

Theorem (Maschke): Let M be a $\mathbb{C}G$-module. Then any submodule has a complement.
Proof: Let $N < M$, $p: M \longrightarrow N$ a $\mathbb{C}$-linear projection. Let $P = e_1 p$. Then
- $n \in N \Rightarrow P(n) = \frac{1}{|G|} \sum_g g \underbrace{p(\underbrace{g^{-1}n}_{\in N})}_{g^{-1}n} = n$,
- $m \in M \Rightarrow P(m) = \frac{1}{|G|} \sum_g g \underbrace{p(g^{-1}m)}_{\in N} \in N$,
- $P^2(m) = P(P(m)) = P(m)$ by the previous two points.

Hence, $P: M \longrightarrow N$ is a $\mathbb{C}G$-linear projection, so $M = \operatorname{Im} P \oplus \operatorname{Ker} P = N \oplus \operatorname{Ker} P$. $\square$

Note: In other words Maschke's theorem states that for $\mathbb{C}G$-modules,
irreducible $\Longleftrightarrow$ indecomposable.
Note: Let $f: M \longrightarrow N$ be a $\mathbb{C}G$-module homomorphism. The complement of its kernel is isomorphic to its image, ie $M \cong \operatorname{Ker} f \oplus \operatorname{Im} f$

Definition: An A-module U is semisimple (completely reducible) if it is isomorphic to a direct sum of simple modules.
Note: According to Maschke's theorem, finite dimensional $\mathbb{C}G$-modules are semisimple.

Note: Let U_i be simple modules, $S < U_1 \oplus \dots \oplus U_n$ a simple submodule. Then $S \cong U_i$ for some i, via the map $p_i \circ i : S \longrightarrow U_i$.

Prop: Let $\mathbb{C}G = U_1 \oplus \dots \oplus U_n$ as a $\mathbb{C}G$-module, where U_i are simple. Let S be any simple $\mathbb{C}G$-module. Then $S \cong U_i$.
Proof: Let $s \in S$. Then $f: \mathbb{C}G \longrightarrow S$, $g \longmapsto s$, is a surjection. Hence, $\mathbb{C}G \cong S \oplus \operatorname{Ker} f$ and S is a submodule of $\mathbb{C}G$.

Note: It follows from Schur's lemma that if U_i are (pairwise non-isomorphic) irreducible $\mathbb{C}G$-modules then
$$\dim \operatorname{Hom}_{\mathbb{C}G}(\oplus U_i^{\oplus a_i}, U_j) = a_j$$
In particular, if $\mathbb{C}G = \oplus U_i^{\oplus a_i}$ then $a_j = \dim U_j$ since
$$a_j = \dim \operatorname{Hom}_{\mathbb{C}G}(\oplus U_i^{\oplus a_i}, U_j) = \dim \operatorname{Hom}_{\mathbb{C}G}(\mathbb{C}G, U_j) = \dim U_j$$
determined on $1 \in \mathbb{C}G$

In particular,
$$|G| = \dim \mathbb{C}G = \sum (\dim U_i)^2$$

Let $V = \operatorname{Sp}_{\mathbb{C}}\{v\}$, $G \longrightarrow \mathbb{C}^\times$ a group homomorphism. Then V becomes a G-module via $g \cdot v = \alpha(g) v$. We denote by $\bar{V}$ the G-module with
$$g \cdot \bar{v} = \overline{\alpha(g)} \bar{v}.$$
Note that
$$g \cdot v \otimes \bar{v} = gv \otimes g\bar{v} = |\alpha(g)|^2 v \otimes \bar{v} = v \otimes \bar{v},$$
g has finite order, so $\alpha(g)$ is a root of unity

so $V \otimes \bar{V} = \mathbb{C}$ is the trivial module. In particular, it follows that $M \otimes V$ is simple if M is.

Definition: Let $\varrho: G \longrightarrow GL(M)$ be a representation. Then its character is the map
$$\chi_M : G \longrightarrow \mathbb{C}, \quad g \longmapsto \operatorname{tr}(\varrho(g)).$$
Note that the degree of χ is $\chi(e) = \dim M$.

Prop: Let χ be a character. Then $\chi(g^{-1}) = \overline{\chi(g)}$.
Proof: For any g, $\varrho(g): M \longrightarrow M$ is diagonalisable (since its minimal polynomial is a product of distinct linear factors). Hence, M has a basis $\{m_1, \dots, m_n\}$ on which
$$\varrho(g) m_i = \lambda_i m_i,$$
where λ_i are roots of unity. Note that $\varrho(g^{-1}) \cdot m_i = \lambda_i^{-1} m_i$. Hence
$$\overline{\chi_M(g)} = \overline{\lambda_1 + \dots + \lambda_n} = \lambda_1^{-1} \oplus \dots \oplus \lambda_n^{-1} = \chi(g^{-1}) \quad \square$$

Prop: Let M, N be $\mathbb{C}G$-modules. Then
1. $\chi_{M \oplus N} = \chi_M \oplus \chi_N$,
2. $\chi_{M^*} = \overline{\chi_M}$,
3. $\chi_{M \otimes N} = \chi_M \chi_N$,
4. $\chi_{\operatorname{Hom}_{\mathbb{C}}(M,N)} = \overline{\chi_M} \chi_N$.

Proof: Fix $g \in G$ and diagonalise $\varrho(g)$.

Definition: A class function on G is a function $f: G \longrightarrow \mathbb{C}$ such that
$$f(xgx^{-1}) = f(g) \ \forall x \in G.$$
The space of class functions becomes an inner product space under
$$\langle \alpha, \beta \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\beta(g)}.$$

Prop: Let M, N be $\mathbb{C}G$-modules. Then
$$\langle \chi_M, \chi_N \rangle = \dim \operatorname{Hom}_{\mathbb{C}G}(M,N).$$
Proof: Let $E: \operatorname{Hom}_{\mathbb{C}}(M,N) \longrightarrow \operatorname{Hom}_{\mathbb{C}}(M,N)$, $\alpha \longmapsto e_1 \cdot \alpha$ be the projection onto $\operatorname{Hom}_{\mathbb{C}G}(M,N)$. Hence, $\dim \operatorname{Hom}_{\mathbb{C}G}(M,N) = \operatorname{Tr} E$. Note that
$$E = \frac{1}{|G|} \sum_{g \in G} \varrho(g),$$
where ϱ is the representation of G on $\operatorname{Hom}_{\mathbb{C}}(M,N)$. Hence,
$$\operatorname{Tr} E = \frac{1}{|G|} \sum_{g \in G} \chi_{\operatorname{Hom}_{\mathbb{C}}(M,N)}(g) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_M(g)} \chi_N(g) = \langle \chi_M, \chi_N \rangle. \quad \square$$

Note: It follows that the irreducible characters are orthonormal. In particular, two representations are isomorphic iff they have the same character.
Note: It also follows that M is simple iff $\langle \chi_M, \chi_M \rangle = 1$.
use Maschke

Note: For class functions α, β,
$$\langle \alpha, \beta \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \beta(g) = \frac{1}{|G|} \sum_{\text{conj. classes}} \overbrace{|g_i^G|}^{\frac{|G|}{|C_G(g_i)|}} \alpha(g_i) \overline{\beta(g_i)} = \sum_{\text{conj. classes}} |C_G(g_i)|^{-1} \alpha(g_i) \overline{\beta(g_i)}.$$

Prop: Let $\{\chi_1, \dots, \chi_n\}$ be the set of irreducible characters of G. Then they form a basis for the space of class functions.
Proof: It suffices to show that $\langle \alpha, \chi_i \rangle = 0 \ \forall i \Rightarrow \alpha = 0$. Let
$$z_\alpha = \sum_{g \in G} \overline{\alpha(g)} g \in Z(\mathbb{C}G).$$
We show that z_α kills any simple module S, and hence also kills $\mathbb{C}G$, a direct sum of simple modules. In particular, $z_\alpha e = z_\alpha = 0$, ie $\overline{\alpha(g)} = 0 \ \forall g$.
Now, since $z_\alpha \in Z(\mathbb{C}G)$,
$$z_\alpha \cdot s = cs \quad \forall s \in S$$
for some $c \in \mathbb{C}$, ie
$$z_\alpha = \sum_{g \in G} \overline{\alpha(g)} \varrho(g) = c\,\mathrm{id}_S.$$
rep. corresponding to S

Hence, taking trace,
$$|G| \overbrace{\langle \chi_S, \alpha \rangle}^{0} = \sum \overline{\alpha(g)} \chi_S(\alpha) = c \dim S.$$
So $c = 0$. $\square$

Theorem: Consider a table

$$\begin{array}{c|c} & \text{conj. classes } g_i \\ \hline \begin{matrix}\text{irr.}\\ \text{characters}\\ \chi_j\end{matrix} & \chi_i(g_j) \end{array}$$

Then

(i) (row orthogonality)
$$\sum_{i=1}^n |C_G(g_i)|^{-1} \chi_r(g_i) \overline{\chi_s(g_i)} = \delta_{rs},$$
(ii) (column orthogonality)
$$\sum_{i=1}^n \chi_i(g_r) \overline{\chi_i(g_s)} = \delta_{rs} |C_G(g_r)|.$$
represent the r^{th} conjugacy class by g_r

Proof: The first part follows from orthonormality of the characters. For the second part, let
$$\delta_s : G \longrightarrow \mathbb{C}, \quad g \longmapsto \begin{cases} 1 & \text{if } g \in s^{\text{th}} \text{ conjugacy class } g_s^G \\ 0 & \text{o/w}. \end{cases}$$
Then
$$\langle \delta_s, \chi_i \rangle = |G|^{-1} \sum_{g \in G} \delta_s(g) \overline{\chi_i(g)} = |G|^{-1} |g_s^G| \overline{\chi_i(g)} = |C_G(g_s)|^{-1} \overline{\chi_i(g)}$$
Hence,
$$\delta_s = \sum_{i=1}^n |C_G(g_s)|^{-1} \overline{\chi_i(g_s)} \chi_i.$$
Now the result follows by evaluating at g_r. $\square$